asymmetric (that is, different impacts for El Niño and La Niña events) from some modes such as the IOB, SPMM and SASD. The impact from the IOB on La Niña SSTA is much more pronounced than on El Niño SSTA, consistent with previous findings[14].

Predictability reduced by model biases

Next, we turn to the impacts of biases in comprehensive climate models on ENSO forecast skill. We conducted more XRO model forecast experiments by using the operator parameters trained using the 91 historical simulation outputs from the Coupled Model Intercomparison Project (CMIP) phase 5 and 6 ('The XRO reforecasting experiments based on CMIP model outputs' section in the Methods and Extended Data Table 2, red curves in Extended Data Fig. 1). Figure 5a reveals that the forecast skill of XRO^m, when trained solely on each CMIP CGCM, shows a wide inter-CGCM spread at a lead time from 7 to 17 months. The forecast skill when the model is trained on CMIP output is consistently lower than for the model trained on observational data (Extended Data Fig. 7a). This suggests that biases in all climate models reduce the ability of these CGCMs to forecast ENSO correctly.

We modified each XRO^m to remove these dynamical biases, by individually substituting the parameters obtained from the observations into three key components of the model: ENSO's internal dynamics ($\mathbf{L}_{\mathrm{ENSO}}$), the remote climate mode feedbacks onto ENSO ($\mathbf{C}_1$) and the ENSO teleconnections to the remote modes ($\mathbf{C}_2$). Correcting the ENSO dynamics ($\mathbf{L}_{\mathrm{ENSO}}$) generally enhances forecast skill at all lead times (red curve in Fig. 5b, Extended Data Fig. 7b). This indicates that the way ENSO's core dynamics are biased in climate models is a main factor in lower ENSO forecast skill. Correcting the remote climate mode feedbacks onto ENSO ($\mathbf{C}_1$) also improves the ENSO forecasts for lead times up to 16 months (magenta curve in Fig. 5b, Extended Data Fig. 7c). Thus, mode coupling is critical for ENSO development as another source of bias. Correcting the ENSO teleconnections ($\mathbf{C}_2$) yields reduced ENSO skill (blue curve in Fig. 5b, Extended Data Fig. 7d), but greatly improves the forecast skill for other modes, such as the IOD (Extended Data Fig. 8). These results indicate that reduced biases in model ENSO dynamics and in climate mode interactions lead to more skilful ENSO forecasts.

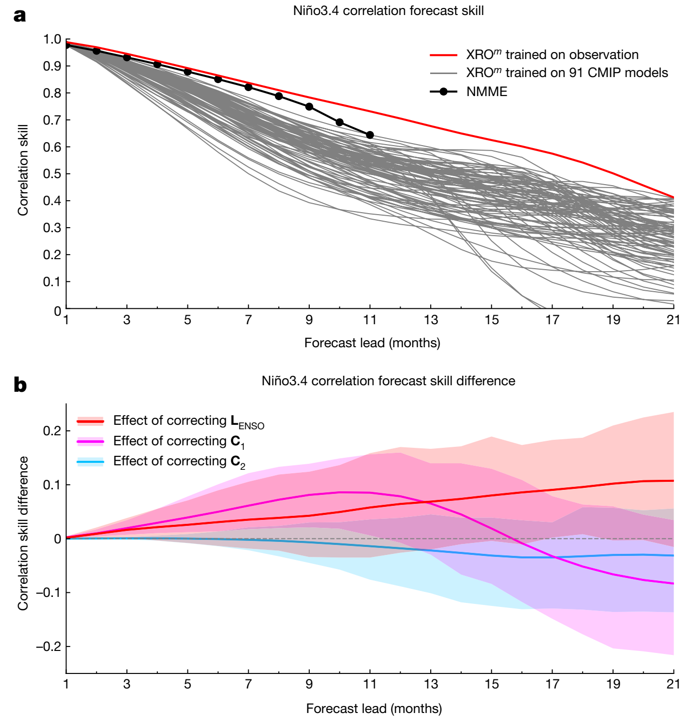

Fig. 5 | Linking biases in the dynamics captured by the XRO to climate model deficiencies in forecasting ENSO during 1979–2022. a, The all-months correlation skill of the 3 month running mean Niño3.4 index in XRO^m trained solely on 91 individual CMIP model outputs (grey curves), and in XRO trained on observations (red) and multi-model ensemble mean NMME models (black). **b**, The ensemble mean and 10–90% spread band of the changes in correlation skill of the Niño3.4 index, obtained by correcting ENSO's internal linear dynamics (XRO$^m_{\mathbf{L}_{\mathrm{ENSO}}}$-XRO^m, red), correcting the remote climate mode feedbacks onto ENSO (XRO$^m_{\mathbf{C}_1}$-XRO^m, magenta) or correcting ENSO's teleconnections to the remote climate modes (XRO$^m_{\mathbf{C}_2}$-XRO^m, blue). Reforecasts using the XRO trained on climate model output show that reduced biases in model ENSO dynamics and in climate mode interactions lead to more skilful ENSO forecasts.

Pantropical SST predictability

Last, we demonstrate that ENSO–climate mode interactions also enhance the SST predictability of other climate modes. For instance, the lead time of skilful IOB forecast extends from 5 months in the uninitialized ENSO experiment to 19 months in the XRO control experiment (Supplementary Fig. 13c,j). The all-month IOD forecast skill extends to 5 months (the September–October–November forecast to 8 months), supporting earlier findings that long-lead IOD predictability arises from ENSO and is affected by the signal-to-noise ratio[45]. The improvement is also evident for SSTA modes in the AO (about 1 month, Supplementary Fig. 13f–h). There is no skill improvement to NPMM and SPMM, possibly because their initial state already includes ENSO information given the strong nearly instantaneous feedback with ENSO[38] (Fig. 2e,f).

In addition to ENSO amplitude, our XRO model can be expanded to also consider ENSO spatiotemporal diversity by using two ENSO SST indices (for example, the Niño3 and Niño4 indices, as in the model XRO2; section 'The XRO2 ENSO types and pantropical SSTA forecasts' in the Methods). The XRO2 is able successfully predict the EP-type characteristic of the 1997–1998 El Niño, and the mixed-type characteristic of the 2015–2016 El Niño, up to 9 months in advance (Supplementary Table 3). By contrast, the North American Multi-Model Ensemble (NMME) dynamical models fail to predict the correct type for the 1997–1998 event, possibly due to long-standing model biases of westward-displaced ENSO SSTAs[46]. The successful prediction of ENSO spatial diversity in the XRO has important implications for predicting global climate impacts that differ strongly for contrasting ENSO SSTA patterns. Furthermore, the skill of forecasted pantropical SSTA at a 9 month lead using the regression model of ten forecasted SST indices outperforms the operational dynamical models in most regions except the Caribbean Sea (Supplementary Fig. 14). The successful forecasts of ENSO types and pantropical SSTA within the XRO framework highlight the essential importance of accurately representing ENSO–climate mode interactions in climate models for effective seasonal forecasting.

Discussion

The XRO model constitutes a parsimonious representation of the climate system in a reduced variable and parameter space that still captures the essential dynamics of interconnected global climate variability. We emphasize that the improvement of ENSO predictability in the XRO relative to that in the nRO ultimately all resides in the initial condition memory of the other climate modes, which is propagated forward by the unbiased operator. Thus, to improve ENSO predictions, climate models must correctly capture the recharge oscillator dynamics of ENSO and, furthermore, three compounding aspects of other climate modes: (1) the initial conditions of each mode, (2) the seasonally modulated damping rate (that is, the memory) of each mode and (3) the seasonally modulated teleconnection to ENSO from each mode. Tracing biases from the SSTA budget at the process level with the XRO framework can be used to inform climate model development. Moreover, the explainable predictability of pantropical climate